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SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

(192262o133)

I Veebala Jeevan *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Concepts	25%	2	1	1	1
Explanation	25%	2	2	1	1
Application	20%	1	1	1	2
Organization	15%	1	1	2	1
Accuracy	10%	1	1	1	2
Clarity	5%	1	2	2	1

32/40

Total Marks /40 *Signature of the Faculty*

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of

transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7.An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8.A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9.A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of

segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

(39) CO1, AT1 (39)

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Computer Networks

1) A Company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 while the Data Link Layer encapsulates each segment 40 bytes of header and trailer.

2) A TransPort Layer Protocol uses a sliding window size of 6 frames Each frame carries 1024 bytes of user data. If the sender transmits 48 frames. calculate the number of transmissions windows required. If one communication

3) A 12,000 byte IP Packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment a 20-byte IP header.

4) A TransPort Layer Protocol using a sliding window size of 6 frames Each frames carrying 1024 bytes of user data.

file Transfer using Transport layer and data Link Layer

Introduction

In computer network data is transferred from one computer to another through different layers of the network model

given data

file size = 150 mB

Network bandwidth = 50 mbps

Segment size = 1500 bytes

Data Link Layer header and trailer = 40 bytes

Per frame

we need to calculate the total number of Segments and frames transmission overhead and total transmission time.

$1 mB = 10^6$ bytes

$150 mB = 150 \times 10^6$

Each segment carries 1500 bytes

the number of segments is calculated as

Number of Segments = $\frac{\text{Total file size}}{\text{Segment size}}$

$= \frac{150,000,000}{1500}$

Data Link Layer

1) converts segments into frames
2) Adds headers and trailers
3) Uses error detection such as CRC to detect corrupted frames
4) supports frame retransmission on link that provide this feature.

final Answer

Number of Segments = 100,000

Number of frames = 100,000

Total overhead = 4MB

Window size = 6 frames

Total frames = 48

Data Per frame = 1024 bytes

One lost frame Per window

formula

Number of windows = Total frames / window size

Calculation

$= \frac{48}{6}$

= 8 windows

Calculate the total data transmitted

formula

Total data = Number of frames x Data Per frame

Calculation

= 48 x 1024

= 49,152 bytes

= 48 KB

Calcate the total number of retrans missions

Number of windows = 8

Total retrans missions = 8x1

= 8 retransmissions

4) Explain flow control

flow control prevents a fast sender from sending more data than the receiver can

Advantages

1) Prevents receiver buffer overflow
2) Controls the rate of data transmission
3) Reduces unnecessary data loss
4) Improves communication efficiency
5) Allows multiple frames to be transmitted before waiting for acknowledgements.

final Answers

Number of windows = 8

Total data = 49,152 bytes

Total retransmissions = 8 frames

Given

Total IP Packet size = 12,000 bytes

MTU = 1500 bytes

IP header size = 20 bytes

Assume the 12,000 byte Packet includes its original IP header

1) calculate the maximum data per fragment

formula

Maximum fragment data = MTU - IP header

Calculation

= 1600 - 20

= 1480 bytes

2) calculate the original data size

Original data size - Total Packet size - original IP header

= 12,000 - 20

= 11,980 bytes.

3) Calculate the number of segments

11,950 / 1460 = 8.09

Therefore, we need 9 segments

= 8 x 1460

= 11,680 bytes

= 11,950 - 11,680

= 140 bytes

4) Calculate the total header overhead

formula

Total header overhead = Number of segments

Header size

Calculation => 9 x 20 = 180 bytes

final answers

Number of segments = 9

Total header overhead = 180 bytes

Total transmitted size = 12,180 bytes.

Given

- window size = 6 frames
- Total frames = 46
- Data Per frame = 1024 bytes
- Last frames Per window = 1

1) Calculate the number of windows

formula

Number of windows = total frames / window size

Calculation

= 46/6

= 8 windows

2) Calculate the total data

formula

Total data = Number of frames * Data Per frame

Calculation

= 46 * 1024

= 49,152 bytes.

Calculate retransmissions

One frame is last in each window

Total retransmissions = Number of windows * Last frames per window.

= 8 * 1

= 8 retransmissions

4) Explain the advantages of flow control

1) Prevents the receiver from becoming overloaded
2) Controls the speed of data transmission
3) Reduces buffer overflow.
4) Improves the use of network resources

final Answer

Number of windows = 8

Total data = 49,152 bytes

Retransmissions = 8 frames

Given

total file size = 600 MB

checkpoint interval = 60 MB

Data transferred before failure = 390 MB

1) calculate the number of checkpoints

formula

Number of checkpoints = Data transferred

Calculation

= 390/60

= 6.5

Six complete checkpoints have been reached before the failure

2) Calculate the data to be retransmitted

Data transmitted after the latest checkpoint

= 390 - 360

= 30 MB

3.) Explain the importance of synchronization

1) synchronization creates checkpoints during data transmission
2) checkpoints help communication resume after a failure
3) The system does not need to restart the entire transfer.
4) It is useful for video streaming file transfer and long communication

final Answers

- Number of completed checkpoints = 6
- Latest checkpoint = 360 MB
- Data to be transmitted = 30 MB.